L4: Representation Learning (Part 2)

Principal Component Analysis & Eigenvectors

Machine Learning Techniques (MLT) Course Notes — Lecture 4 Summary

1. Real-World Reality & Motivation

In Lecture 3, we assumed that points lay perfectly on a given line and calculated projections for off-line points. In real-world applications, however, two fundamental challenges arise:

1. **Exact Linear Relationships Are a Myth:** Real-world data (such as height vs. weight measurements) almost never lie perfectly on a line due to noise, measurement error, and underlying stochasticity.
2. **No One Gives Us the Line:** In practice, no external supervisor provides the direction vector w or identifies which points are "on" or "off" the line. We must discover the best line directly from raw, unlabelled data.

Compression Ratio vs. Reconstruction Error: In $\mathbb{R}^2$, **any** straight line passing through the origin provides the exact same **50**% compression ratio (storing 1 direction vector $w \in \mathbb{R}^d$ plus n scalar coordinates $c_i \in \mathbb{R}$ instead of $2n$ raw numbers). However, different candidate lines yield drastically different **reconstruction errors** when mapping projected points back into feature space.

Figure 1: Choosing the Best Representative Line in Real-World

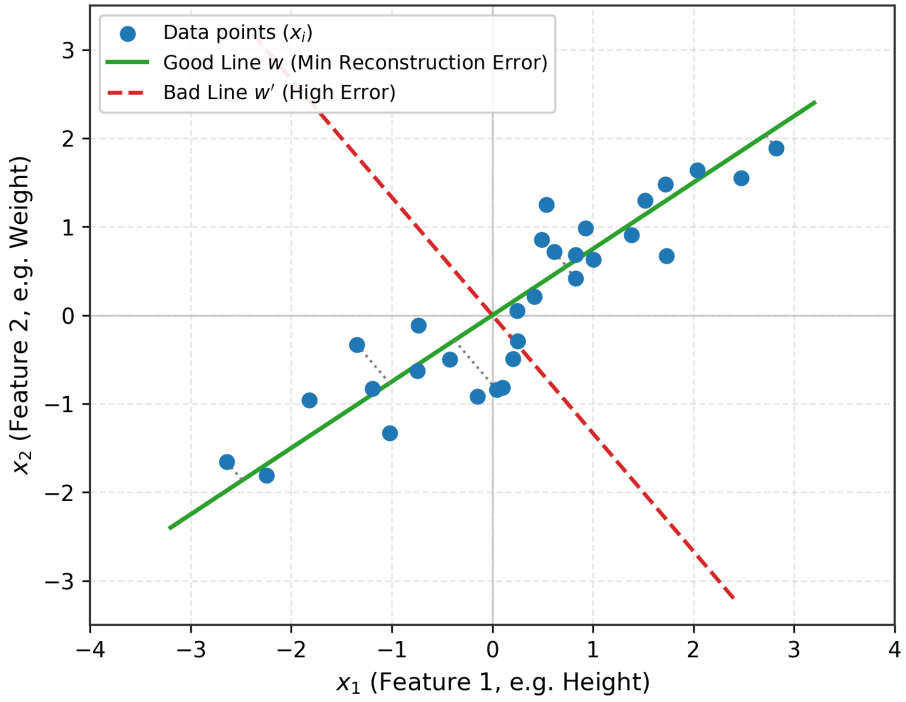


Figure 1: Comparing candidate lines through the origin. Line w minimizes reconstruction error, whereas line w' results in high error.

Core Objective: Given an unlabelled dataset $\{x_1, x_2, \ldots, x_n\}$ where $x_i \in \mathbb{R}^d$, find the direction vector $w \in \mathbb{R}^d$ (with $\|w\|^2 = 1$) that **minimizes the total (or average) reconstruction error**.

2. Mathematical Formulation of Reconstruction Error

Let $w \in \mathbb{R}^d$ be a candidate unit direction vector satisfying $\|w\|^2 = w^T w = 1$.

- The **scalar coordinate** (projection length) of point x_i along line w is $c_i = x_i^T w$.
- The **proxy vector** (reconstructed point in $\mathbb{R}^d$) is $\hat{x}_i = c_i w = (x_i^T w)w$.
- The **reconstruction error vector** is $e_i = x_i - (x_i^T w)w$.
- The **squared error distance** for point x_i is $\|x_i - (x_i^T w)w\|^2$.

Averaging across all n data points defines the **Average Reconstruction Error Function** $f(w)$:

$$f(w) = \frac{1}{n} \sum_{i=1}^{n} \|x_i - (x_i^T w)w\|^2 \tag{1}$$

Figure 2: Orthogonal Projection & Error Vector Decomposition

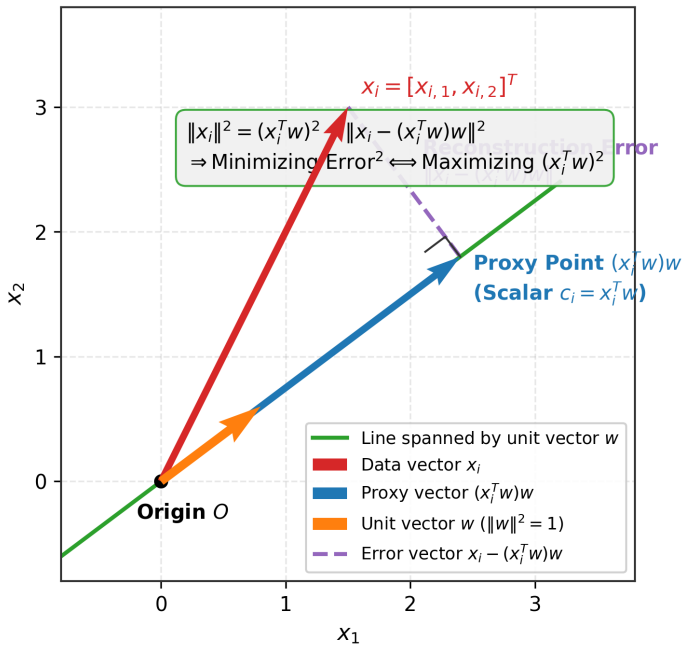


Figure 2: Geometric decomposition of point x_i into its proxy vector $(x_i^T w)w$ and perpendicular error vector.

3. Step-by-Step Algebraic Expansion

Using the vector norm identity $\|z\|^2 = z^T z$ for any vector $z \in \mathbb{R}^d$:

$$\|x_i - (x_i^T w)w\|^2 = \big(x_i - (x_i^T w)w\big)^T \big(x_i - (x_i^T w)w\big) \tag{2}$$

Expanding term-by-term:

$$= x_i^T x_i - x_i^T (x_i^T w)w - \big((x_i^T w)w\big)^T x_i + \big((x_i^T w)w\big)^T \big((x_i^T w)w\big) \tag{3}$$

Since $(x_i^T w)$ is a scalar quantity, we re-order scalar factors:

- $x_i^T (x_i^T w)w = (x_i^T w)(x_i^T w) = (x_i^T w)^2$

- $\big((x_i^T w)w\big)^T x_i = (x_i^T w)(w^T x_i) = (x_i^T w)^2$
- $\big((x_i^T w)w\big)^T \big((x_i^T w)w\big) = (x_i^T w)^2 (w^T w)$

Substitute these back and apply the unit length constraint $\|w\|^2 = w^T w = 1$:

$$\|x_i - (x_i^T w)w\|^2 = \|x_i\|^2 - 2(x_i^T w)^2 + (x_i^T w)^2(1) = \|x_i\|^2 - (x_i^T w)^2 \tag{4}$$

Substituting this simplified squared error back into the average error function $f(w)$:

$$f(w) = \frac{1}{n}\sum_{i=1}^{n} \|x_i\|^2 - \frac{1}{n}\sum_{i=1}^{n} (x_i^T w)^2 \tag{5}$$

4. Converting Minimization to Maximization

Notice that the first term in Equation (5), $\frac{1}{n}\sum_{i=1}^{n} \|x_i\|^2$, is the **average squared magnitude of the original raw dataset**. This term is a constant with respect to w.

Therefore, **minimizing reconstruction error** $f(w)$ is mathematically equivalent to **maximizing the second term**:

$$\max_{w:\|w\|^2=1} g(w) = \frac{1}{n}\sum_{i=1}^{n} (x_i^T w)^2 \tag{6}$$

Geometric Intuition: $(x_i^T w)^2$ represents the variance of the data projected onto line w. Minimizing reconstruction error is identical to maximizing the variance of the projected data!

5. Matrix Form & The Covariance Matrix C

Since $(x_i^T w)$ is a scalar, we write its square as a product:

$$(x_i^T w)^2 = (w^T x_i)(x_i^T w) = w^T (x_i x_i^T) w \tag{7}$$

Plugging Equation (7) back into $g(w)$ and pulling w outside the summation:

$$g(w) = \frac{1}{n}\sum_{i=1}^{n} w^T (x_i x_i^T) w = w^T \left(\frac{1}{n}\sum_{i=1}^{n} x_i x_i^T\right) w \tag{8}$$

We define the $d \times d$ **Covariance Matrix** C:

$$C = \frac{1}{n}\sum_{i=1}^{n} x_i x_i^T \quad \in \mathbb{R}^{d\times d} \tag{9}$$

Thus, the optimization problem reduces to maximizing the bilinear form:

$$\max_{w:\|w\|^2=1} w^T C w \tag{10}$$

Core Concept & Solution Summary

We started discussing about finding the line, and the line is the one corresponding to the vector w that maximizes $w^T C w$ subject to $\|w\|^2 = 1$:

$$\max_{w:\|w\|^2=1} w^T C w \quad \text{where } C = \frac{1}{n}\sum_{i=1}^{n} x_i x_i^T$$

From the Rayleigh Quotient Theorem in linear algebra, the solution w^* to this constrained optimization problem is the **eigenvector corresponding to the largest eigenvalue** ($\lambda_{\max}$) of the covariance matrix C of the data points:

$$Cw^* = \lambda_{\max} w^* \quad \text{with } \|w^*\|^2 = 1$$

This optimal vector w^* defines the **First Principal Component** of the dataset—the direction that simultaneously minimizes average reconstruction error and maximizes projected data variance.